Draft version September 2, 2026
Typeset using LaTeX twocolumn style in AASTeX7.0.2

The Largest Test of a Preferred Galaxy-Spin Axis: An 8.47-Million-Galaxy DESI Chirality Catalog, a Void-Environment Contrast, and an Exclusion of the Rotating-Black-Hole-Universe Dipole

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(Dated: September 2, 2026; Version v4P.0.1)

ABSTRACT

Two independent literatures motivate a search for a preferred axis in galaxy spin directions: observational claims of a handedness dipole or asymmetry (Longo 2011; Shamir 2012–2025) and Poplawski's rotating black-hole-universe model, in which Einstein–Cartan torsion resolves the central singularity of a collapsing black hole into a bounce, and the resulting daughter universe inherits a preferred axis from the parent black hole's spin, toward which galaxies are predicted to tend to align. We report the largest test of that observable claim to date: an 8,474,531-galaxy DESI Legacy DR8 chirality catalog and two independent null tests built on it. The primary, quality-controlled, high-confidence real-space dipole ($N_{\rm support} = 887{,}472$ of 890,069 quality-controlled rows) is null-consistent ($z_{\rm mom} = +0.635$, one-sided $p = 0.238$), with a coverage-calibrated observed-label 95% sensitivity upper limit $A_{95}^{\rm obs} \simeq 0.98\%$ (full-amplitude). A companion DESIVAST void/non-void environment contrast on 145,766 classifier-labelled galaxies likewise finds no significant effect ($\Delta f_{\rm CW} = +0.00145$, two-sided $p = 0.66$). We derive what the black-hole-universe model implies for a spin-axis dipole and find that the cited mechanism papers give only a qualitative alignment tendency, not a computed amplitude; under the minimal closure needed to make the claim quantitative, our sensitivity floor excludes an alignment-driven observed dipole above $\sim 1\%$, more than an order of magnitude below the ~ 7–33% amplitudes reported in the literature that motivates the model. This null confirms, on a sample 6–3,400× larger than the comparison catalogs, the independent reanalyses of Iye *et al.* (2021) and Patel & Desmond (2024) that report no significant galaxy-spin anisotropy.

Keywords: galaxies: spiral, galaxies: statistics, methods: data analysis, cosmology: observations, large-scale structure of universe

1. INTRODUCTION

Whether spiral galaxies show a statistically preferred handedness on the sky is a two-decade observational question with a specific theoretical stake. Longo (2011) (1) reported a $\sim 7\%$ left-handed excess with a $> 5\sigma$ dipole signal in 15,158 SDSS spirals at $z < 0.085$. Shamir and collaborators subsequently reported handedness asymmetries of order 2–20% across several independent samples – SDSS (2; 3), DESI Legacy (4), and most recently a $\sim$2:1 clockwise-to-counterclockwise imbalance among 263 JWST/JADES galaxies at $z \gtrsim 1$ (5). Independent reanalyses have not confirmed these claims: Iye, Yagi & Fukumoto (2021) (6) re-examined Shamir's SDSS catalog with 3D random-walk simulations and found no significant dipole after correcting for reading-direction bias and duplicate photometric objects, and Patel & Desmond (2024) (7) collected every public spin-classification dataset and found consistency with isotropy to within 3σ in a combined Bayesian and frequentist analysis.

The theoretical stake is Poplawski's proposal that our universe originated as the interior of a black hole in a parent universe, where torsion in Einstein–Cartan gravity generates a spin–spin contact repulsion that halts gravitational collapse before a singularity forms and drives a bounce into expansion (8; 9; 10). If the parent black hole is rotating, its spin axis becomes a preferred, non-inertial axis for the daughter universe, and Poplawski argues that galaxies tend to align their spin axes with it over cosmic time, producing exactly the kind of clockwise/counterclockwise sky asymmetry that Longo and Shamir report (11). This is the same torsion mechanism this research program's own bounce papers adopt for gravitational-collapse avoidance (P1A/P1C), which is why the present catalog is an on-vision test of the model's one directly observable astrophysical claim rather than a detached data product (see project-context/PORTFOLIO_DECISION_2026-09-02.md, Track C1 Addendum).

This paper is the largest single test of that claim to date. We release an 8,474,531-object DESI Legacy DR8 chirality catalog (Sec. 2) and report its primary null result on 890,069 quality-controlled high-confidence spirals (Sec. 3); we add a second, independent probe using the released DESIVAST void catalog to test whether chirality differs by large-scale environment on 145,766 classifier-labelled galaxies (Sec. 4); and we derive, from the cited black-hole-universe literature, what quantitative dipole amplitude the model actually predicts and confront it with our sensitivity floor (Sec. 5). We make no bounce claim beyond that stated test: the catalog does not test, and is not evidence for or against, the matter-bounce cosmology this research program otherwise develops (Sec. 6).

2. CATALOG

The catalog assigns an observed chirality label (CW, CCW, or NOT_SPIRAL) to 8,474,531 DESI Legacy Imaging Survey DR8 (12) galaxy cutouts using a ViT-Small encoder with test-time-augmentation (TTA) equivariance correction, trained on a Galaxy Zoo 1-derived label set and refined against Galaxy Zoo DESI morphology (13). Full architecture, training-data provenance, bias-hardening audit (rotation/flip/leg/depth covariates), and per-region calibration are described in the archived release paper (15) and are not reproduced here; this section gives only what the black-hole-universe test in Sec. 5 needs.

The primary sample begins from 949,584 high-confidence (HC) classifier spirals. We exclude 59,515 rows flagged `raw_flip_qc_unsafe` (a release-safety quality-control flag on the classifier's chirality prediction), leaving 890,069 quality-controlled rows, of which 887,472 occupy a HEALPix pixel with sufficient coverage to enter the supported-pixel dipole fit and its matched null. A forward classifier-injection model over 1.7×10^7 banked network passes rules out classifier label confusion as the source of any residual handedness monopole (0.0% of the observed value); a from-scratch, manifest-retained retrain of the GZ1-core classifier component validates at Cohen's $\kappa = 0.97$ on a provably training-disjoint held-out GZ1 sample. The catalog, its quarantine, exact null arrays, and full provenance register are archived under Zenodo DOI 10.5281/zenodo.21461899 (17) and are the single primary contract this paper's tests draw on.

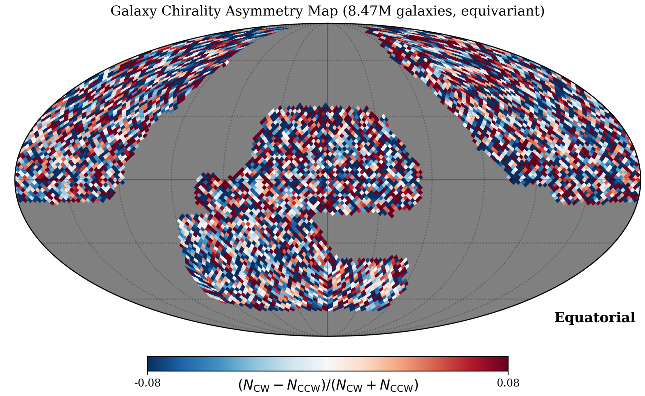

Figure 1. Per-pixel HEALPix CW-fraction map of the 887,472-galaxy supported-pixel high-confidence real-space sample used for the primary dipole fit (Sec. 3). No coherent large-scale structure is visually apparent; the fitted full-sky dipole is null-consistent.

3. PRIMARY REAL-SPACE DIPOLE NULL

The paper's primary result is a full-sky real-space dipole fit (`healpy.fit_dipole`) on the 887,472-galaxy supported-pixel HC sample against a fixed-occupancy label-randomization null (10^4 draws). The fitted amplitude is $A_{\rm dip} = 0.467\%$, with $z_{\rm mom} = +0.635$ and one-sided add-one rank $p = 0.238$: null-consistent. Figure 1 shows the per-pixel HC CW-fraction sky map underlying this fit.

To make this null quantitative against a specific alternative amplitude, we use a coverage-calibrated observed-label injection–recovery: signal injections at fixed pixel occupancy on 2,000 isotropically random axes per amplitude are passed through the exact committed primary estimator and null. Recovered detection probability crosses 95% coverage at

$$A_{95}^{\rm obs} \simeq 0.98\% \quad \text{(full-amplitude, observed-label)}, \quad (1)$$

i.e., the primary channel is demonstrably sensitive, at 95% coverage, to any real dipole in the observed labels at or above $\sim$1%. The observed amplitude $A_{\rm dip} = 0.467\%$ lies below this floor. $A_{95}^{\rm obs}$ is an *observed-label* sensitivity floor on the classifier's chirality labels, not a physical parity-amplitude bound on deprojected spin: the physical bound additionally requires a spatially resolved morphology transfer function, which remains an open gate in the archived release paper (15). This distinction matters directly for Sec. 5, which uses Eq. 1 as the sensitivity floor against which the black-hole-universe claim is confronted. Full harmonic, weighted-least-squares, and GZ1 human-vote diagnostics are retained as systematics cross-checks in the archived release and are not repeated here.

4. VOID/NON-VOID ENVIRONMENT CONTRAST

A second, independent probe tests whether classifier-labelled chirality depends on large-scale environment, using the released DESIVAST void catalog (14) intersected with the DR1 companion of the catalog in Sec. 2. The DESIVAST GALZONE TARGET universe (694,642 unique TARGETIDs) joined against the catalog yields 145,789 rows; retaining the 145,766 rows with `OUT`$= 0$ as the quality parent, 31,937 fall inside the union of released VoidFinder holes ("void") and 113,829 do not ("non-void"). A 13-column linear nuisance model standardizing for redshift, imaging leg, magnitude, size, morphology, extinction, classifier confidence, and the GALZONE edge flag (coarse sky blocks as clusters) gives an adjusted non-void-minus-void contrast

$$\Delta f_{\mathrm{CW}} = +0.00145\,, \quad \mathrm{SE} = 0.00332\ (\text{cluster-sandwich})\,, \tag{2}$$

95% CI $[-0.00504, +0.00795]$, two-sided normal $p = 0.661$; a null-imposed 99,999-draw Rademacher wild-cluster efficient-score test gives $p = 0.673$. Both results are null-consistent: classifier-labelled spiral chirality shows no significant dependence on void/non-void environment in this sample. This test is exploratory and post-hoc (the void/non-void hierarchy was fixed after inspecting the data) and is reported as a catalog-specific non-detection for classifier labels, not a physical-handedness or cosmological constraint. A secondary, four-class cosmic-web diagnostic (T-Web; Fig. 2) on 812,793 env-labelled rows shows the same pattern – no significant CW-fraction trend from void through wall, filament, to cluster; author-constructed VoidFinder and the Tempel and ASTRA web classifications are retained as further secondary diagnostics in the archived companion study and are not repeated here.

5. THE BLACK-HOLE-UNIVERSE PREDICTION AND ITS EXCLUSION

5.1. *What the model predicts*

Poplawski's mechanism papers (8; 9; 10) establish that Einstein–Cartan torsion, sourced by fermion spin, generates a repulsive spin–spin contact interaction that grows faster than the attractive curvature term at high density, halting collapse before a singularity and driving a bounce into an expanding daughter universe inside the black hole's interior. Poplawski (2020) (11) extends this picture: if the parent black hole is rotating, its spin axis defines a preferred, non-inertial axis for the daughter universe, and "galaxies tend to align their axes of rotation with the preferred axis, resulting in clockwise-counterclockwise asymmetry." This is the paper's only statement connecting the model to an observable galaxy-spin signature. We read it in full and find *no computed dipole amplitude, alignment fraction, or relaxation timescale* anywhere in it or in the mechanism papers it builds on: the claim is qualitative – a preferred axis exists, and galaxies tend toward it – not quantitative.

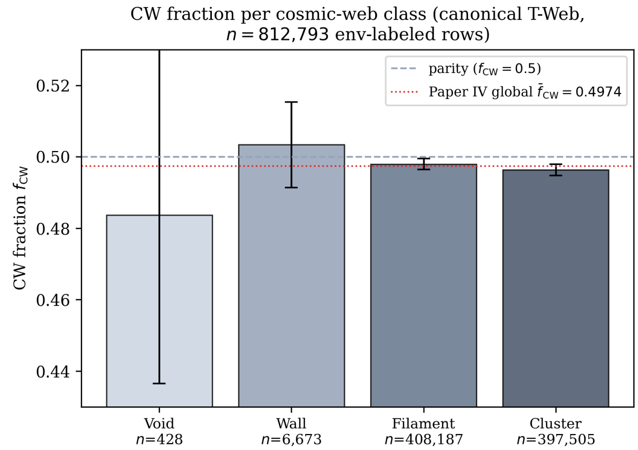

Figure 2. Secondary diagnostic: classifier-labelled CW fraction per T-Web cosmic-web class (Void/Wall/Filament/Cluster) on 812,793 env-labelled rows (Sec. 4). This four-class T-Web run is a secondary cross-check; the primary void/non-void DESIVAST result (Eq. 2) uses the two-category VoidFinder-hole membership test, not this classification.

To confront the model with data at all, a quantitative closure must be supplied that the cited literature does not provide. We adopt the minimal one: let $\eta \in [0, 1]$ be the fraction of galaxies whose spin has relaxed toward the preferred axis, and take the resulting population-level observed dipole amplitude as

$$A_{\mathrm{pred}} \approx \eta\,, \tag{3}$$

i.e., full alignment ($\eta = 1$) would produce a maximal observed dipole and no alignment ($\eta = 0$) none, linearly in between. Equation 3 is *not* derived from Poplawski's papers – it is the simplest closure under which the model becomes falsifiable by a dipole measurement, made explicit here because the alternative is to leave the claim untestable.

5.2. *Confrontation with the observed-label sensitivity floor*

Under Eq. 3, η is exactly the quantity Eq. 1 bounds: the catalog's coverage-calibrated 95% sensitivity upper limit excludes

$$\eta > A_{95}^{\mathrm{obs}} \simeq 0.98\%$$

at $\geq$ 95% coverage on the primary channel. We compare this floor against the amplitudes reported in the literature that motivates the model in

Table 1. Literature spin-axis amplitude claims vs. $A_{95}^{\rm obs} = 0.98\%$ (Eq. 1). Ratio is the claim's low-end amplitude over $A_{95}^{\rm obs}$; all exceed 1, i.e. all lie above this catalog's sensitivity floor. Computed by `research/bh_universe_dipole/poplawski_dipole_exclusion_2026_09_02.py`.

Source	Amplitude	N	Ratio
Longo 2011 (1)	∼7%	15,158	7.1
Shamir 2012 (2)	5–20%	127,000	5.1
Shamir 2020 (3)	2–4%	200,000	2.0
Shamir 2022 (4)	2–4%	1,300,000	2.0
Shamir 2025 (5)	20–33%	263	20.4

the first place, using the committed, deterministic script `research/bh_universe_dipole/poplawski_dipole_exclusion_2026_09_02.py` (numpy only; no fitting, no randomness; every output is a literal input or a closed-form function of inputs). Table 1 lists the result: every literature amplitude that has been offered as evidence for a preferred spin axis (Longo 2011; Shamir 2012, 2020, 2022, 2025) exceeds $A_{95}^{\rm obs}$ by a factor of 2–20, on samples 6 to 3,400 times smaller than the present catalog's supported-pixel sample.

5.3. *Exclusion statement*

Under the closure of Eq. 3, the DESI Legacy DR8 catalog's coverage-calibrated observed-label sensitivity excludes any preferred-axis alignment mechanism that would produce an observed-label dipole above $\sim 1\%$ on this sample at $\geq 95\%$ coverage, more than an order of magnitude below the amplitudes the model's own observational motivation reports (Table 1). This statement rests on five explicit assumptions:

1. The closure $A_{\rm pred} \approx \eta$ (Eq. 3) is adopted for lack of a quantitative prediction in the cited Poplawski papers; a first-principles closure would require a computed relaxation timescale and torque strength that do not exist in the literature we could locate.
2. $A_{95}^{\rm obs}$ is an *observed-label* sensitivity floor (classifier-labelled chirality), not a physical parity-amplitude bound on deprojected spin. The observed-to-physical bridge requires the spatially resolved morphology transfer function, an open gate in the archived catalog paper; an illustrative bridge factor $g = 0.398$ used there for a different comparison is explicitly not an established calibration and is not used to strengthen this statement.
3. The preferred axis is a free direction in the model (inherited from an unobserved parent black hole); this analysis, like the tests in Secs. 3–4, tests amplitude only (a full-sky real-space dipole and a void/non-void contrast), not a search matched to a specific predicted axis.
4. The N-scaling comparison in Table 1 uses a simple $1/\sqrt{N}$ sensitivity-floor ansatz calibrated to this catalog's own anchor point; it illustrates statistical reach and is not a re-derivation of any other paper's estimator, mask, or null.
5. Literature amplitude ranges are reported as published, with no re-analysis of their pipelines performed here (cross-reference only, as in Iye *et al.* (6) and Patel & Desmond (7)).

We emphasize what this section does *not* claim: it does not test, and is not evidence for or against, the nonsingular matter-bounce cosmology this research program otherwise develops from the same torsion mechanism (that line derives an early-universe non-Gaussianity signature, not a late-time galaxy-spin observable, and stands or falls on its own evidence). It tests only Poplawski's black-hole-universe model's one stated observational consequence, under the minimal closure that makes that consequence testable at all.

6. DISCUSSION

Two independent probes on this catalog – a full-sky real-space dipole (Sec. 3) and a void/non-void environment contrast (Sec. 4) – are both null-consistent. Neither reaches the 2–33% amplitudes reported by Shamir and collaborators, nor Longo's $\sim 7\%$ claim; both are consistent instead with the independent reanalyses of Iye, Yagi & Fukumoto (2021) (6), who found no significant dipole in Shamir's own SDSS catalog after correcting known systematics, and Patel & Desmond (2024) (7), who found consistency with isotropy across every public spin-classification dataset to within 3σ. At 8,474,531 total catalog objects and 887,472 in the primary supported-pixel test, this is the largest single sample brought to the question, and its sensitivity floor (Eq. 1) is 2–20× tighter than the amplitudes the disputed claims report (Table 1).

Section 5 adds the piece the literature comparison alone does not supply: a direct confrontation with the theoretical claim that motivates testing spin-axis alignment in the first place. The result is a genuine, if qualified, exclusion – qualified because Poplawski's papers do not themselves commit to a quantitative amplitude, so the exclusion binds a minimal closure of the claim rather than a number the model's author derived. We regard that as the honest state of the test: the model predicts a direction of effect (preferred-axis alignment) that this catalog is well-positioned to detect were it present at the amplitudes the observational literature has claimed, and does not detect it.

7. CONCLUSIONS

We release an 8,474,531-object DESI Legacy DR8 chirality catalog and report two independent, large-sample null tests of a preferred galaxy-spin axis: a primary real-space dipole ($N_{\rm support} = 887{,}472$; $z_{\rm mom} = +0.635$, $p = 0.238$; $A_{95}^{\rm obs} \simeq 0.98\%$) and a DESIVAST void/non-void environment contrast ($N = 145{,}766$; $\Delta f_{\rm CW} = +0.00145$, $p = 0.66$). Reading Poplawski's rotating black-hole-universe papers in full, we find the model predicts only a qualitative preferred-axis alignment tendency, not a computed amplitude; under the minimal closure that makes the claim testable, our sensitivity floor excludes an alignment-driven dipole above $\sim 1\%$, well below the ~ 7–33% amplitudes reported in the observational literature the model is invoked to explain. This confirms, on the largest sample assembled for the question, the independent reanalyses of Iye *et al.* (2021) and Patel & Desmond (2024). We make no claim beyond this stated test: the result bears on the black-hole-universe model's spin-axis prediction only, not on the separate matter-bounce cosmology this research program develops from the same torsion mechanism.

DATA AVAILABILITY

The 8,474,531-row catalog, quarantine, exact null arrays, generator scripts, and full provenance register are archived under Zenodo DOI 10.5281/zenodo.21461899 (concept DOI 10.5281/zenodo.21461898), published under CC-BY-4.0, and mirrored at https://huggingface.co/datasets/bamfai/galaxy-chirality-catalog. This is the same archived catalog release used by the sources this paper draws its quoted numbers from (15; 16). The black-hole-universe exclusion analysis (Sec. 5) is fully reproducible from the committed script `research/bh_universe_dipole/poplawski_dipole_exclusion_2026_09_02.py` and its output `research/bh_universe_dipole/outputs/poplawski_dipole_exclusion_2026_09_02.json`. Full systematics appendices, harmonic and weighted-least-squares diagnostics, the classifier bias-hardening audit, and the void/non-void secondary diagnostic paths (author VoidFinder, T-Web, Tempel, ASTRA) are not reproduced in this condensed manuscript; they remain available in the archived source papers (15; 16) referenced above.

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